Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

2.1 – Speed — Measuring It Ourselves

Unit 2: Forces and Motion

Two numbers, one word

Before today, **speed** is a word you already use — a car is fast, a snail is slow. In physics it is something more useful: a word that stands for **two measurements put together**.

Every speed, however it is reported, comes from exactly two things:

how far and **how long it took**

Today you measure both of those yourself, for your own walking and running, before a single formula appears on this page. The formula is just a short way of writing down something you will have already worked out by hand.

The practical

Measure your own speed

Kit: trundle wheel, stopwatch (one per pair), two marker cones.

1. As a class, use the trundle wheel to mark out a straight **20 m** track. Put a cone at each end.
2. Work in pairs. One partner moves along the track; the other times with the stopwatch — start it the instant your partner crosses the first cone, stop it the instant they cross the second.
3. Cover the same 20 m four times, once at each pace: a **normal walk**, a **fast walk**, a **jog**, and your **fastest sprint possible**.
4. Swap roles and repeat, so both partners finish with a full set of four times.
5. Record your own four times in the table overleaf.

Care: several pairs share the track at once — agree a direction with the group next to you before you start timing, so nobody meets someone coming the other way. For the sprint, run straight through the second cone and keep clear space beyond it to slow down in.

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Your results

Fill in **Distance** and **Time** for all four paces now. Leave the **Speed** column blank — you will come back to it in Section 3.

Pace	Distance / m	Time / s	Speed / m/s
Normal walk			
Fast walk			
Jog			
Fastest sprint possible			

Naming what we measured

You already have everything you need to work out how fast each pace was: a distance and a time. Physics has a short way of writing the calculation you are about to do:

speed = distance ÷ time
distance in **metres (m)**, time in **seconds (s)**, speed in **metres per second (m/s)**

Go back to your table and work out the speed for each of your four paces. Write each answer in the Speed column, to 1 decimal place.

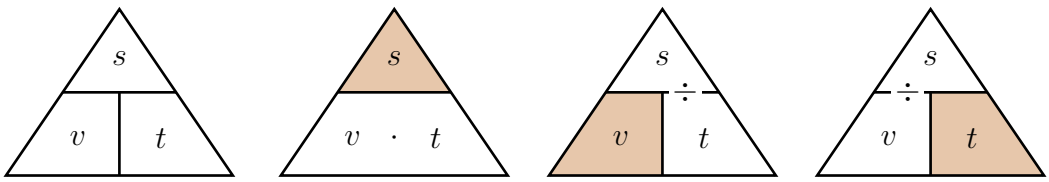
Check: for the same 20 m, your sprint speed should come out as the *largest* number and your normal walk as the *smallest*. A faster pace means a shorter time for the same distance, which means dividing by a smaller number — so the answer gets bigger.

Triangle math

Physicists shorten the words to letters: **distance is** s, **speed is** v and **time is** t. So speed = distance ÷ time becomes

$$v = s \div t$$

Put the letters in the triangle. **Shade the one you are looking for**, and what is left tells you what to do: **one above the other means divide**, **side by side means multiply**.



the triangle **shade** s**:** $s = v \cdot t$ **shade** v**:** $v = s \div t$ **shade** t**:** $t = s \div v$

Check: shade v and you get back the formula you just used on your table.

Conclusions

1) **State** the formula for speed, with its units.

> **State:** Give a specific name, value or other brief answer without explanation or calculation.

Model answer: Speed is distance divided by time: speed (m / s) = distance (m) ÷ time (s).

2) **Describe** the pattern between your pace and your speed.

> **Describe:** Give a detailed account or picture of a situation, event, pattern or process.

Model answer: As the pace gets faster — from normal walk to fast walk to jog to sprint — the time taken to cover the same 20 m gets shorter, so the calculated speed gets larger each time.

3) **Explain** why two people who both run the *same* 20 m race can finish with different speeds.

> **Explain:** Give a detailed account including reasons or causes.

Model answer: Speed depends on both distance and time. If the distance is fixed at 20 m for both runners, the only thing that can differ between them is the time — whoever takes less time to cover the same distance has divided 20 m by a smaller number, giving a larger speed.

Extension

Predict, then test. Your jog speed is in metres per second. Without doing the practical again, work out roughly how far you would jog in one *minute* at that speed, and how fast that is in kilometres per hour. Then check your working against a partner's.

Multiply the jog speed (m / s) by 60 for metres per minute, then by 60 again and divide by 1000 for km / h — equivalently, multiply the original m / s figure by 3.6.

Criterion A — Knowing and understanding

This task is **formative**. It does not go on your report — we are *practising* the writing, and this is the standard we are practising against. Your three answers above (**State**, **Describe**, **Explain**) are the evidence.

Read down the right-hand column and notice what changes: the **verb** gets more demanding each time, and the situations go from **familiar** to **unfamiliar**.

Level	The student is able to:
0	Does not reach a standard indicated by any of the descriptors below.
1–2	i. **recall** scientific knowledge ii. apply scientific knowledge and understanding to **suggest solutions** to problems set in **familiar** situations iii. **apply** information to make judgments.
3–4	i. **state** scientific knowledge ii. apply scientific knowledge and understanding to **solve** problems set in **familiar** situations iii. **apply** information to make **scientifically supported** judgments.
5–6	i. **outline** scientific knowledge ii. apply scientific knowledge and understanding to solve problems set in familiar situations **and suggest solutions to problems set in unfamiliar situations** iii. **interpret** information to make scientifically supported judgments.
7–8	i. **describe** scientific knowledge ii. apply scientific knowledge and understanding to solve problems set in **familiar and unfamiliar** situations iii. **analyse** information to make scientifically supported judgments.

Wording taken verbatim from the *MYP Sciences guide* (2015), assessment criteria **Year 3**, Criterion A. Bold emphasis added to show what changes between levels.